



Vectors

V₈



Vectors

↓ Random

↓ Eigen

$$\begin{bmatrix} 2 \times d \end{bmatrix}$$



Vectors

=



Vectors

Both Magnitude
and
Direction changes

Ex:

Matrix $\Rightarrow \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$



Vector $\Rightarrow \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

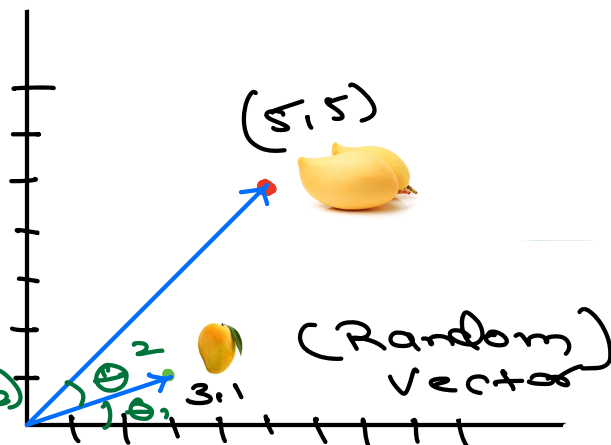
Θ_1



= np.dot(M, )

$$\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

(Θ_1, Θ_2)



$$\begin{bmatrix} 2 \times d \end{bmatrix}$$



Vectors

=

λ



Vectors

Eigen $\vec{V}$

Ex:



Vector $\Rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$\Rightarrow \Theta_1$



Vector $\Rightarrow \text{np.dot}(M, \text{Mentos Fruit logo})$

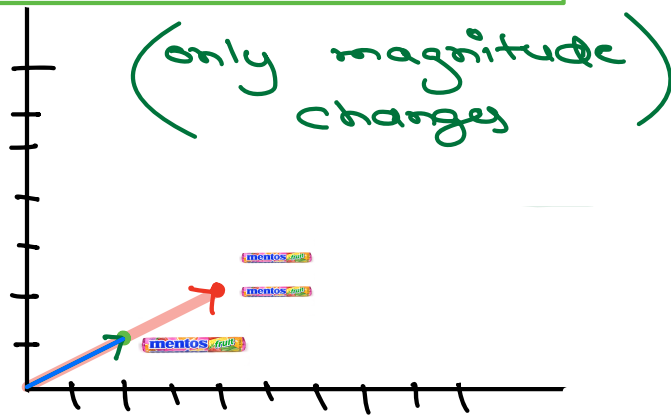


$$\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$\Rightarrow \Theta_1$

$$\Rightarrow \Theta_1$$



$(2, 1) \rightarrow 2$
Eigen Value
Eigen Value

Note that $M \cdot \vec{V} = \lambda \cdot \vec{V}$ where $\lambda = 2$

Q: How do we find  Vector for any matrix

→ $M\vec{v} = \lambda\vec{v}$ can be rewritten as

→ $M\vec{v} - \lambda\vec{v} = 0$

→ $(M - \lambda I)\vec{v} = 0$

↘ we know that $\vec{v} \neq 0$
Hence we can solve for
 $\det(M - \lambda I) = 0$
and
find values of λ

→ Using λ 's we can find $\vec{v}$

Ex:

$$M = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$$

Solving

$$\det(M - \lambda I) = 0$$

$$\text{i.e. } \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \det \begin{bmatrix} 1-\lambda & 2 \\ 3 & -4-\lambda \end{bmatrix}$$

$$\Rightarrow (1-\lambda) \cdot (-4-\lambda) - 2 \cdot 3$$

$$\Rightarrow -4-\lambda + 4\lambda + \lambda^2 - 6$$

$$\Rightarrow \lambda^2 + 3\lambda - 10$$

$$\Rightarrow (\lambda - 2)(\lambda + 5) = 0$$

Hence $\lambda_1 = 2$ and $\lambda_2 = 5$ (Eigen Values)

To Do: Now using λ find $\vec{v}$ (Eigen Vectors)

